# Econometrics II – Exam

Spring 2026

## 1 Part 3 - The Additional Assignment

Throughout, we work with the VAR(1) model (5.1)–(5.3) and, for Questions (3)–(7), the modified specification (5.5)–(5.7). The stationarity assumptions $|\phi_1| < 1$ and $|\phi_3| < 1$ are maintained.

### Question (1) – Reduced-form impulse responses

**(a) Closed form for $\phi_1 \neq \phi_3$.** We show by induction on $k \geq 1$ that

$$\Phi^k = \begin{pmatrix} \phi_1^k & \phi_2 \, \dfrac{\phi_1^k - \phi_3^k}{\phi_1 - \phi_3} \\ 0 & \phi_3^k \end{pmatrix}. \tag{1}$$

*Base case* $k = 1$. Directly, $\Phi_{12}^{(1)} = \phi_2 = \phi_2(\phi_1 - \phi_3)/(\phi_1 - \phi_3)$, and the diagonal entries are $\phi_1, \phi_3$, as required.

*Inductive step.* Assume (1) holds for some $k \geq 1$. Because $\Phi$ is upper-triangular the product $\Phi^{k+1} = \Phi \, \Phi^k$ remains upper-triangular with diagonal $(\phi_1^{k+1}, \phi_3^{k+1})$, and the $(1, 2)$-element is

$$\begin{aligned} \Phi_{12}^{(k+1)} &= \phi_1 \cdot \Phi_{12}^{(k)} + \phi_2 \cdot \Phi_{22}^{(k)} = \phi_1 \, \phi_2 \, \frac{\phi_1^k - \phi_3^k}{\phi_1 - \phi_3} + \phi_2 \, \phi_3^k \\ &= \frac{\phi_2}{\phi_1 - \phi_3} \Big( \phi_1^{k+1} - \phi_1 \phi_3^k + (\phi_1 - \phi_3) \phi_3^k \Big) = \phi_2 \, \frac{\phi_1^{k+1} - \phi_3^{k+1}}{\phi_1 - \phi_3}. \end{aligned}$$

This completes the induction and establishes (5.4).

**(b) Special case $\phi_1 = \phi_3$.** When the two diagonal entries coincide the geometric-difference quotient in (1) is of indeterminate form $0/0$. Treating $\phi_3$ as the variable and applying L'Hôpital's rule,

$$\lim_{\phi_3 \to \phi_1} \frac{\phi_1^k - \phi_3^k}{\phi_1 - \phi_3} = \lim_{\phi_3 \to \phi_1} \frac{-k \phi_3^{k-1}}{-1} = k \, \phi_1^{k-1},$$

so that

$$\Phi_{12}^{(k)}\Big|_{\phi_1=\phi_3} = \phi_2\, k\, \phi_1^{k-1}, \qquad k \geq 1. \tag{2}$$

The same expression is obtained directly by induction: $\Phi_{12}^{(k+1)} = \phi_1\, \phi_2 k \phi_1^{k-1} + \phi_2 \phi_1^k = \phi_2(k+1)\phi_1^k$.

**(c) Cumulated impulse response, $\phi_1 \neq \phi_3$.** Note $\Phi^0 = I$, so $\Phi_{12}^{(0)} = 0$. For $|\phi_1| < 1$ and $|\phi_3| < 1$ the two geometric series below converge, and

$$\begin{aligned}
\sum_{k=0}^{\infty} \Phi_{12}^{(k)} &= \sum_{k=1}^{\infty} \phi_2 \frac{\phi_1^k - \phi_3^k}{\phi_1 - \phi_3} = \frac{\phi_2}{\phi_1 - \phi_3}\left(\sum_{k=1}^{\infty} \phi_1^k - \sum_{k=1}^{\infty} \phi_3^k\right) \\
&= \frac{\phi_2}{\phi_1 - \phi_3}\left(\frac{\phi_1}{1-\phi_1} - \frac{\phi_3}{1-\phi_3}\right) = \frac{\phi_2}{\phi_1 - \phi_3} \cdot \frac{\phi_1 - \phi_3}{(1-\phi_1)(1-\phi_3)}.
\end{aligned}$$

Hence

$$\sum_{k=0}^{\infty} \Phi_{12}^{(k)} = \frac{\phi_2}{(1-\phi_1)(1-\phi_3)}. \tag{3}$$

*Interpretation.* This is the $(1,2)$-element of $(I - \Phi)^{-1} = \sum_{k=0}^{\infty} \Phi^k$ (which converges since both eigenvalues of $\Phi$ lie strictly inside the unit circle), and it measures the long-run effect on $y$ of a unit innovation in $\varepsilon_{2,t}$.

## Question (2) – OLS consistency under conditional heteroskedasticity

We now suppose $\varepsilon_t = H_t^{1/2} u_t$ with $u_t$ i.i.d., $\mathbb{E}[u_t] = 0$, $\mathrm{Var}(u_t) = I_2$, and $H_t$ a positive-definite, $\mathcal{I}_{t-1}$-measurable conditional covariance matrix. The process $\{\varepsilon_t\}$ is assumed stationary.

**Claim.** The equation-by-equation OLS estimator of $\Phi$ is *consistent*.

**Argument.** Writing the VAR(1) compactly as $Z_t = \Phi Z_{t-1} + \varepsilon_t$, the OLS estimator (on the system, equivalent equation-by-equation since the regressors are identical across equations) satisfies

$$\widehat{\Phi}_{\mathrm{OLS}} - \Phi = \left(\frac{1}{T}\sum_{t=1}^{T} \varepsilon_t Z_{t-1}'\right)\left(\frac{1}{T}\sum_{t=1}^{T} Z_{t-1} Z_{t-1}'\right)^{-1}.$$

Two conditions secure consistency:

*(i) Orthogonality.* Because $H_t^{1/2} \in \mathcal{I}_{t-1}$ and $u_t$ is i.i.d. and independent of $\mathcal{I}_{t-1}$,

$$\mathbb{E}[\varepsilon_t \mid \mathcal{I}_{t-1}] = H_t^{1/2}\, \mathbb{E}[u_t] = 0,$$

so by the law of iterated expectations $\mathbb{E}[\varepsilon_t Z'_{t-1}] = \mathbb{E}\big[Z'_{t-1}\mathbb{E}[\varepsilon_t \mid \mathcal{I}_{t-1}]\big] = 0$. The unconditional moment condition that drives OLS is therefore preserved.

*(ii) LLN/ergodic theorem.* Stationarity of $\{(Z_t, \varepsilon_t)\}$ together with weak dependence (inherited from $|\phi_1| < 1$, $|\phi_3| < 1$ and the i.i.d. structure of $u_t$) ensures

$$\frac{1}{T}\sum_{t=1}^{T} Z_{t-1}Z'_{t-1} \xrightarrow{p} \mathbb{E}[Z_{t-1}Z'_{t-1}], \qquad \frac{1}{T}\sum_{t=1}^{T} \varepsilon_t Z'_{t-1} \xrightarrow{p} 0,$$

the first limit being positive definite (otherwise some linear combination of $Z_{t-1}$ would be degenerate, contradicting stationarity with non-zero innovation variance). The continuous mapping theorem then yields $\widehat{\Phi}_{\text{OLS}} \xrightarrow{p} \Phi$.

**Caveats.** Although consistent, OLS is no longer efficient: the Gauss–Markov / MLE-equals-OLS argument relied on $\mathbb{E}[\varepsilon_t \varepsilon'_t \mid \mathcal{I}_{t-1}] = \Sigma$ being constant, which now fails. Two practical implications: (1) standard errors must be made heteroskedasticity-consistent (sandwich/HCSE form), and (2) an estimator that models $H_t$ explicitly (e.g. multivariate GARCH-based quasi-MLE) is asymptotically more efficient.

## Question (3) – Unconditional variance in the ARCH-X model

We work with the modified model (5.5)–(5.7). Assuming a stationary solution exists, $\mathbb{E}[\varepsilon_t^2]$ is constant in $t$. By the law of iterated expectations,

$$\mathbb{E}[\varepsilon_t^2] = \mathbb{E}\big[\mathbb{E}[\varepsilon_t^2 \mid \mathcal{I}_{t-1}]\big] = \mathbb{E}[\sigma_t^2] = \omega + \alpha\,\mathbb{E}[\varepsilon_{t-1}^2] + \gamma\,\mathbb{E}[x_{t-1}^2].$$

The process $\{x_t\}$ is a stationary Gaussian AR(1) with $|\phi_3| < 1$, $\mathbb{E}[x_t] = 0$, and

$$\mathbb{E}[x_t^2] = \frac{\sigma_\eta^2}{1-\phi_3^2}. \tag{4}$$

Imposing stationarity $\mathbb{E}[\varepsilon_t^2] = \mathbb{E}[\varepsilon_{t-1}^2]$ and solving,

$$\mathbb{E}[\varepsilon_t^2] = \frac{\omega + \gamma\,\sigma_\eta^2/(1-\phi_3^2)}{1-\alpha}, \tag{5}$$

which is (5.8).

**Conditions for** (5) **to be well-defined and strictly positive.**

$$\omega > 0, \quad \alpha \geq 0, \quad \gamma \geq 0, \quad \sigma_\eta^2 > 0, \quad |\phi_3| < 1, \quad \alpha < 1.$$

Under these, the numerator of (5) is the sum of two non-negative terms with $\omega > 0$, hence strictly positive, and the denominator $1-\alpha$ is strictly positive. (The strict inequality $\alpha < 1$ is what is actually needed; positivity of $\omega$ and of $\sigma_\eta^2$ ensures the variance is strictly positive even at the boundary $\gamma = 0$.)

**Comparison with standard ARCH(1).** Setting $\gamma = 0$ reduces (5) to the familiar ARCH(1) unconditional variance $\omega/(1-\alpha)$ (cf. §2 of the ARCH lecture notes). The additional term $\gamma\sigma_\eta^2/[(1-\alpha)(1-\phi_3^2)]$ captures the contribution of the persistent exogenous regressor $x_{t-1}^2$ to the unconditional variance of $\varepsilon_t$: a more persistent $x_t$ (larger $\phi_3^2$) or a more variable $\eta_t$ (larger $\sigma_\eta^2$) inflates $\mathbb{E}[\varepsilon_t^2]$, and the effect is amplified by the ARCH feedback through the factor $1/(1-\alpha)$.

## Question (4) – Conditional joint density and log-likelihood

**Factorization.** Conditional on $\mathcal{I}_{t-1}$, the quantities $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma x_{t-1}^2$ and $\phi_3 x_{t-1}$ are constants (deterministic functions of the conditioning information and $\psi$). Since $z_t$ and $\eta_t$ are mutually independent and independent of $\mathcal{I}_{t-1}$, the pair $(y_t, x_t) = (\phi_1 y_{t-1} + \phi_2 x_{t-1} + \sigma_t z_t,\ \phi_3 x_{t-1} + \eta_t)$ is, given $\mathcal{I}_{t-1}$, a function of two independent random variables. Hence

$$f(y_t, x_t \mid \mathcal{I}_{t-1}; \psi) = f(y_t \mid \mathcal{I}_{t-1}; \psi)\, f(x_t \mid \mathcal{I}_{t-1}; \psi). \tag{6}$$

With $z_t \sim N(0,1)$ and $\eta_t \sim N(0, \sigma_\eta^2)$,

$$f(y_t \mid \mathcal{I}_{t-1}; \psi) = \frac{1}{\sqrt{2\pi\,\sigma_t^2}} \exp\left(-\frac{(y_t - \phi_1 y_{t-1} - \phi_2 x_{t-1})^2}{2\sigma_t^2}\right), \tag{7}$$

$$f(x_t \mid \mathcal{I}_{t-1}; \psi) = \frac{1}{\sqrt{2\pi\,\sigma_\eta^2}} \exp\left(-\frac{(x_t - \phi_3 x_{t-1})^2}{2\sigma_\eta^2}\right). \tag{8}$$

**Log-likelihood.** Treating $(y_0, x_0, \varepsilon_0)$ as given initial conditions, the conditional log-likelihood for the sample $\{(y_t, x_t)\}_{t=1}^T$ is, by (6),

$$\begin{aligned} \ell_T(\psi) &= \sum_{t=1}^T \log f(y_t \mid \mathcal{I}_{t-1}; \psi) + \sum_{t=1}^T \log f(x_t \mid \mathcal{I}_{t-1}; \psi) \\ &\equiv \ell_T^y(\theta_y) + \ell_T^x(\theta_x), \end{aligned} \tag{9}$$

where $\theta_y = (\phi_1, \phi_2, \omega, \alpha, \gamma)'$ and $\theta_x = (\phi_3, \sigma_\eta^2)'$, with

$$\ell_T^y(\theta_y) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\log\sigma_t^2 - \frac{1}{2}\sum_{t=1}^{T}\frac{(y_t - \phi_1 y_{t-1} - \phi_2 x_{t-1})^2}{\sigma_t^2}, \tag{10}$$

$$\ell_T^x(\theta_x) = -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\sigma_\eta^2 - \frac{1}{2\sigma_\eta^2}\sum_{t=1}^{T}(x_t - \phi_3 x_{t-1})^2. \tag{11}$$

**Can the $y$- and $x$-equation parameters be estimated separately?** Yes. The decomposition (9) shows that $\ell_T^y$ depends only on $\theta_y$ and $\ell_T^x$ only on $\theta_x$, while the parameter spaces of $\theta_y$ and $\theta_x$ are variation-free (no cross-restrictions). Hence the joint maximum-likelihood estimator factorises:

$$\widehat{\psi} = \big(\widehat{\theta}_y, \widehat{\theta}_x\big), \quad \widehat{\theta}_y = \arg\max_{\theta_y} \ell_T^y(\theta_y), \quad \widehat{\theta}_x = \arg\max_{\theta_x} \ell_T^x(\theta_x).$$

This is the standard cut argument of the lecture notes on conditional models: under independence of the components and variation-free parameter spaces, the marginal model carries no information about $\theta_y$ and the conditional model carries no information about $\theta_x$, so equation-by-equation estimation is fully efficient (in particular, $\widehat{\theta}_x$ is just the OLS estimator of the AR(1) for $x_t$, and $\widehat{\theta}_y$ is the (Q)MLE of the conditionally-Gaussian ARCH-X model for $y_t$).

## Question (5) – Multi-step conditional variance forecasts

**(a) One- and two-step forecasts.** Since $\sigma_{T+1}^2 = \omega + \alpha\varepsilon_T^2 + \gamma x_T^2$ is $\mathcal{I}_T$-measurable,

$$\sigma_{T+1|T}^2 = \omega + \alpha\,\varepsilon_T^2 + \gamma\,x_T^2. \tag{12}$$

For the two-step forecast, take conditional expectations of $\sigma_{T+2}^2 = \omega + \alpha\varepsilon_{T+1}^2 + \gamma x_{T+1}^2$ given $\mathcal{I}_T$. Using $\varepsilon_{T+1} = \sigma_{T+1} z_{T+1}$ with $z_{T+1}$ independent of $\mathcal{I}_T$ and $\mathbb{E}[z_{T+1}^2] = 1$,

$$\mathbb{E}[\varepsilon_{T+1}^2 \mid \mathcal{I}_T] = \mathbb{E}[\sigma_{T+1}^2 \mid \mathcal{I}_T] \cdot \mathbb{E}[z_{T+1}^2] = \sigma_{T+1|T}^2.$$

For the $x$-term, $x_{T+1} = \phi_3 x_T + \eta_{T+1}$ with $\eta_{T+1}$ independent of $\mathcal{I}_T$, so

$$\mathbb{E}[x_{T+1}^2 \mid \mathcal{I}_T] = (\phi_3 x_T)^2 + \mathrm{Var}(\eta_{T+1}) = \phi_3^2 x_T^2 + \sigma_\eta^2.$$

Combining,

$$\sigma_{T+2|T}^2 = \omega + \alpha\,\sigma_{T+1|T}^2 + \gamma\big(\phi_3^2 x_T^2 + \sigma_\eta^2\big). \tag{13}$$

**(b) Convergence as $h \to \infty$.** For $h \geq 2$ the same argument yields the recursion

$$\sigma^2_{T+h|T} = \omega + \alpha\, \sigma^2_{T+h-1|T} + \gamma\, \mathbb{E}[x^2_{T+h-1} \mid \mathcal{I}_T]. \tag{14}$$

Since $\{x_t\}$ is a stationary AR(1), iterating $x_{T+h-1} = \phi_3^{h-1} x_T + \sum_{j=0}^{h-2} \phi_3^j \eta_{T+h-1-j}$ gives

$$\mathbb{E}[x^2_{T+h-1} \mid \mathcal{I}_T] = \phi_3^{2(h-1)} x_T^2 + \sigma_\eta^2 \sum_{j=0}^{h-2} \phi_3^{2j} \xrightarrow{h\to\infty} \frac{\sigma_\eta^2}{1-\phi_3^2},$$

because $|\phi_3| < 1$. Substituting this limit into (14), the recursion converges (as $h \to \infty$) to the fixed-point equation

$$\sigma^2_\infty = \omega + \alpha\, \sigma^2_\infty + \gamma\, \frac{\sigma_\eta^2}{1-\phi_3^2} \iff \sigma^2_\infty = \frac{\omega + \gamma\sigma_\eta^2/(1-\phi_3^2)}{1-\alpha} = \mathbb{E}[\varepsilon_t^2],$$

by (5). Since $0 \leq \alpha < 1$, the linear recursion (14) is a contraction with rate $\alpha$, so convergence is exponential. (The transient through $\mathbb{E}[x^2_{T+h-1} \mid \mathcal{I}_T]$ decays at the additional rate $\phi_3^2$.)

## Question (6) – GMM under non-Gaussian $z_t$

Suppose now that $z_t$ is i.i.d. with $\mathbb{E}[z_t] = 0$ and $\mathbb{E}[z_t^2] = 1$, but otherwise unrestricted in distribution. The processes $\{z_t\}$ and $\{\eta_t\}$ are independent of each other and of $\mathcal{I}_{t-1}$.

**Step 1: Conditional moment restrictions.** At the true parameter $\psi_0$, since $\sigma_t \in \mathcal{I}_{t-1}$ and $z_t \perp \mathcal{I}_{t-1}$,

$$\mathbb{E}[\varepsilon_t \mid \mathcal{I}_{t-1}] = \sigma_t\, \mathbb{E}[z_t \mid \mathcal{I}_{t-1}] = \sigma_t \cdot 0 = 0, \tag{15}$$

$$\mathbb{E}[\varepsilon_t^2 - \sigma_t^2 \mid \mathcal{I}_{t-1}] = \sigma_t^2\big(\mathbb{E}[z_t^2 \mid \mathcal{I}_{t-1}] - 1\big) = \sigma_t^2\,(1-1) = 0, \tag{16}$$

which are the restrictions in (5.9).

**Step 2: From conditional to unconditional moments via instruments.** Any vector $w_t$ that is $\mathcal{I}_{t-1}$-measurable yields a valid unconditional moment by the law of iterated expectations:

$$\mathbb{E}[\varepsilon_t \cdot w_t] = \mathbb{E}\big[w_t\, \mathbb{E}[\varepsilon_t \mid \mathcal{I}_{t-1}]\big] = 0, \quad \mathbb{E}[(\varepsilon_t^2 - \sigma_t^2) \cdot w_t] = 0.$$

The $y$-equation has five parameters $\theta_y = (\phi_1, \phi_2, \omega, \alpha, \gamma)'$, so we need at least five linearly independent moment functions. A natural just-identified choice is:

$$g_t^{(\text{mean})}(\theta_y) = \varepsilon_t(\theta_y) \begin{pmatrix} y_{t-1} \\ x_{t-1} \end{pmatrix} \in \mathbb{R}^2, \tag{17}$$

$$g_t^{(\text{var})}(\theta_y) = \big(\varepsilon_t(\theta_y)^2 - \sigma_t^2(\theta_y)\big) \begin{pmatrix} 1 \\ \varepsilon_{t-1}(\theta_y)^2 \\ x_{t-1}^2 \end{pmatrix} \in \mathbb{R}^3, \tag{18}$$

where $\varepsilon_t(\theta_y) = y_t - \phi_1 y_{t-1} - \phi_2 x_{t-1}$ and $\sigma_t^2(\theta_y) = \omega + \alpha\varepsilon_{t-1}(\theta_y)^2 + \gamma x_{t-1}^2$. All five components are $\mathcal{I}_{t-1}$-measurable instruments multiplied by either $\varepsilon_t$ or $\varepsilon_t^2 - \sigma_t^2$, so (15)–(16) imply $\mathbb{E}[g_t(\theta_{y,0})] = 0$ with

$$g_t(\theta_y) = \big(g_t^{(\text{mean})}(\theta_y)',\, g_t^{(\text{var})}(\theta_y)'\big)' \in \mathbb{R}^5.$$

The instruments $\{1, \varepsilon_{t-1}^2, x_{t-1}^2\}$ in (18) are the canonical choices for identifying $(\omega, \alpha, \gamma)$, mirroring the regressors of the ARCH-X variance equation; the instruments $\{y_{t-1}, x_{t-1}\}$ in (17) mirror those of the mean equation.

**Step 3: GMM estimator.** Let $\bar{g}_T(\theta_y) = T^{-1}\sum_{t=1}^T g_t(\theta_y)$ and $W_T$ a sequence of positive-definite weight matrices with $W_T \xrightarrow{p} W$. The GMM estimator is

$$\widehat{\theta}_y^{\text{GMM}} = \arg\min_{\theta_y}\; T\,\bar{g}_T(\theta_y)'\, W_T\, \bar{g}_T(\theta_y). \tag{19}$$

With $\dim g_t = \dim \theta_y = 5$ the model is exactly identified, so the criterion attains zero at $\widehat{\theta}_y^{\text{GMM}}$ and $W_T$ is irrelevant for the point estimator. Under standard regularity (identification: $\mathbb{E}[g_t(\theta_y)] = 0$ iff $\theta_y = \theta_{y,0}$; stationarity and weak dependence; differentiability and existence of moments),

$$\sqrt{T}\big(\widehat{\theta}_y^{\text{GMM}} - \theta_{y,0}\big) \xrightarrow{d} N\big(0,\, (D'WD)^{-1}D'WSWD(D'WD)^{-1}\big),$$

with $D = \mathbb{E}[\partial g_t(\theta_{y,0})/\partial\theta_y']$ and $S = \sum_{j=-\infty}^{\infty}\mathbb{E}[g_t(\theta_{y,0})g_{t-j}(\theta_{y,0})']$. In the present setting the conditional moments (15)–(16) imply that $\{g_t(\theta_{y,0})\}$ is a martingale difference sequence with respect to $\{\mathcal{I}_t\}$, so $S = \mathbb{E}[g_t(\theta_{y,0})g_t(\theta_{y,0})']$ (no autocovariance terms) and consistent estimation of $S$ does not require HAC corrections.

**Remark.** Augmenting (17)–(18) with additional lagged instruments produces an over-identified system; in that case the efficient choice $W_T = \widehat{S}^{-1}$ minimises the asymptotic variance and yields Hansen's $J$-test of the over-identifying restrictions. Either way, no assumption on $z_t$ beyond $\mathbb{E}[z_t] = 0$ and $\mathbb{E}[z_t^2] = 1$ is needed.

## Question (7) – Testing $H_0 : \gamma = 0$

By assumption, $\sqrt{T}(\widehat{\theta}_y^{\text{GMM}} - \theta_{y,0}) \xrightarrow{d} N(0, V)$ for some positive-definite $V$, and a consistent estimator $\widehat{V} \xrightarrow{p} V$ is available from the GMM output (the sandwich formula in Step 3 above, which simplifies under the m.d.s. property to $V = (D'S^{-1}D)^{-1}$ for the efficient estimator).

**Wald test.** Let $R = (0, 0, 0, 0, 1) \in \mathbb{R}^{1\times 5}$ select $\gamma$ from $\theta_y$, so that $R\theta_y = \gamma$. Under $H_0 : \gamma = 0$, $\sqrt{T}\, R\, \widehat{\theta}_y^{\text{GMM}} \xrightarrow{d} N(0, RVR') = N(0, V_{\gamma\gamma})$ and

$$\xi_W \;=\; T \left(R\, \widehat{\theta}_y^{\text{GMM}}\right)' \left(R\, \widehat{V}\, R'\right)^{-1} \left(R\, \widehat{\theta}_y^{\text{GMM}}\right) \;=\; T\, \frac{\widehat{\gamma}^{\,2}}{\widehat{V}_{\gamma\gamma}} \;\xrightarrow{d}\; \chi^2(1). \tag{20}$$

Equivalently, the $t$-statistic $t_\gamma = \sqrt{T}\, \widehat{\gamma} / \sqrt{\widehat{V}_{\gamma\gamma}} \xrightarrow{d} N(0, 1)$. Reject $H_0$ at level $\alpha^*$ if $\xi_W > \chi^2_{1-\alpha^*}(1)$ (or, two-sidedly, $|t_\gamma| > z_{1-\alpha^*/2}$).

**Boundary remark.** Strictly speaking, $\gamma \geq 0$ is a maintained restriction of the model, so $H_0 : \gamma = 0$ places the parameter on the boundary of $\Theta$. The asymptotically valid distribution of the one-sided likelihood-ratio analogue is then a $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ mixture (cf. the standard boundary result for testing $H_0 : \alpha = 0$ in the ARCH model). Using the standard $\chi^2(1)$ critical values for the two-sided Wald (20) is conservative in this case.